

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA5612

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.*
(BL4)

Assessment Tool Number	: CO2_AT2
Assessment Tool Weightage	: 35%
Assessment Tool	: Simulation Task
Duration	: Self Learning
Marks	: 100 Marks
Type of Assessment Conduction	: At Home
Evaluation Type	: Online Evaluation(GitHub)
Result Generation	: Students

successfully simulate the assigned wireless communication system, analyze the obtained results, and interpret the performance based on relevant technical parameters.

Topic Covered : LTE/LTE-Advanced architecture, protocol stack, carrier aggregation, higher-order MIMO, beamforming, and beamsteering techniques.
5G NR air interface, gNodeB and RAN architecture, RRC procedures, 5G Core, Service-Based Architecture, network slicing, AMF, SMF, UPF, and control/user plane separation.

Conduction of Assessment Procedure:

- Assessment will be carried out through rubric-based evaluation, where students complete the assigned task.
- Simulation will be done in MATLAB/Simulink for the given specification and interpret the results.
- TWO questions will be assigned to students each carrying 10 marks and they are required to collaboratively discuss the given problem and present their responses with clear justification and submit the document in GitHub Platform.

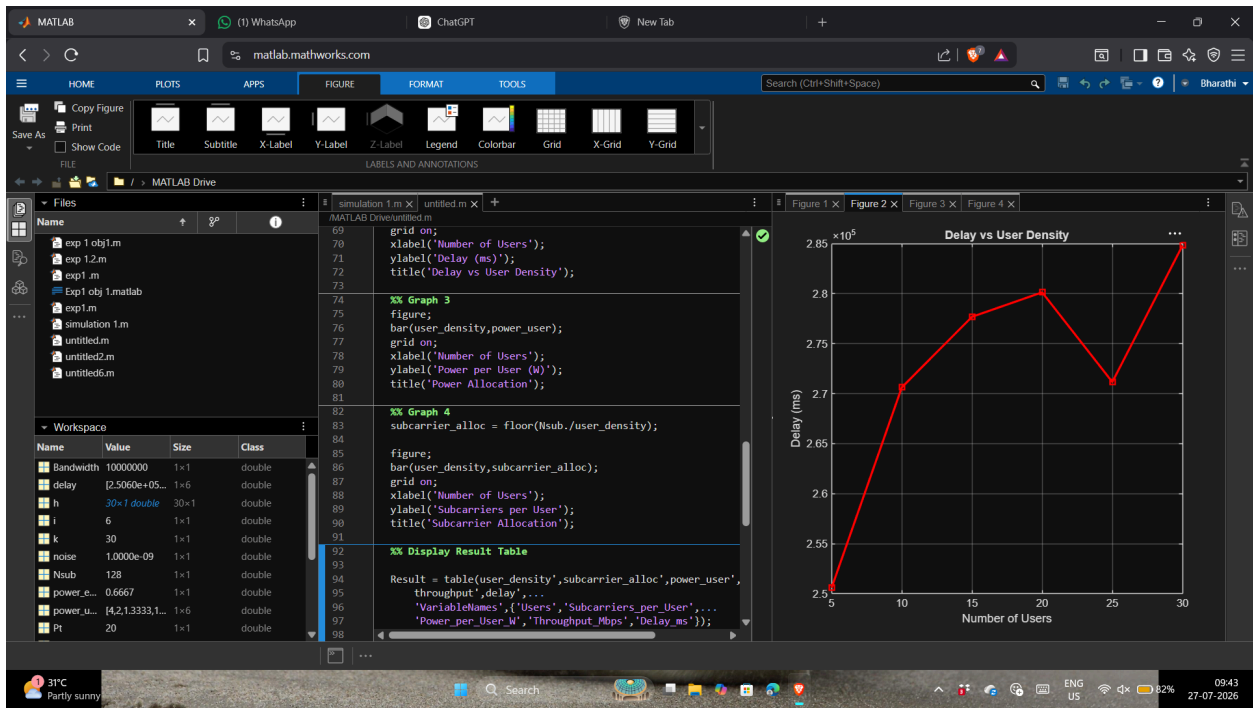
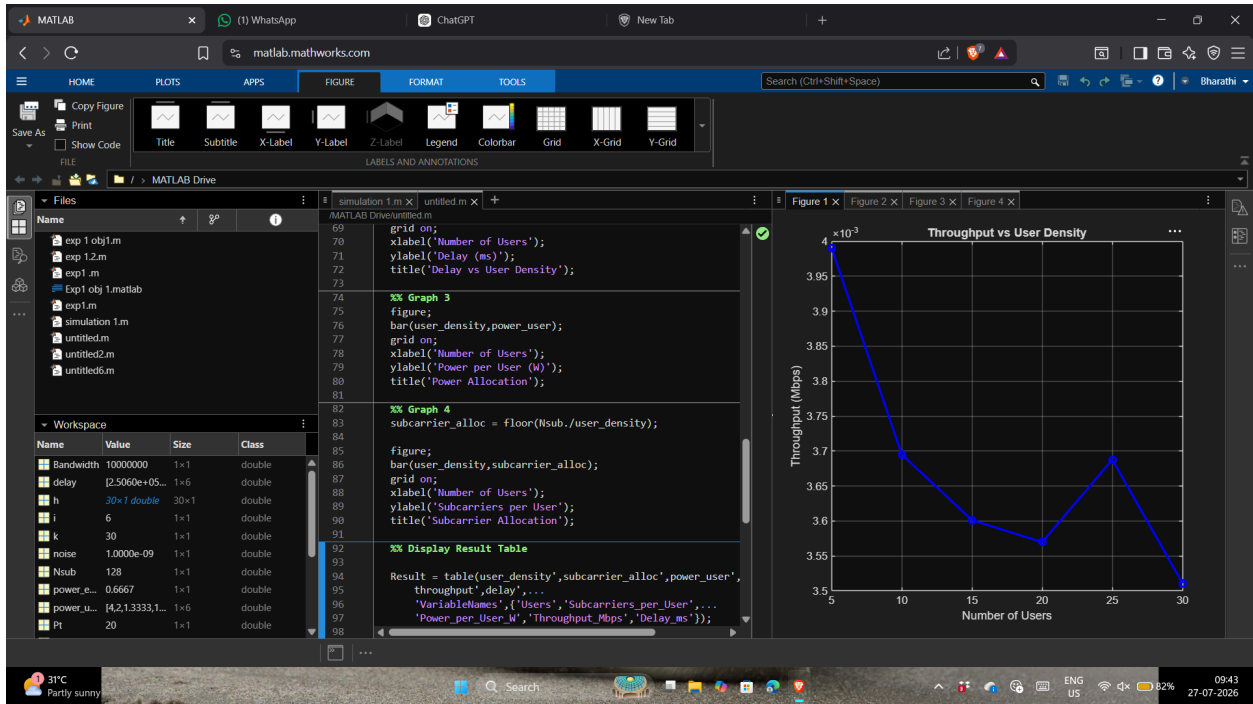
19. Simulation of OFDMA resource allocation under user density, subcarriers, power allocation, throughput, and delay (BL4)

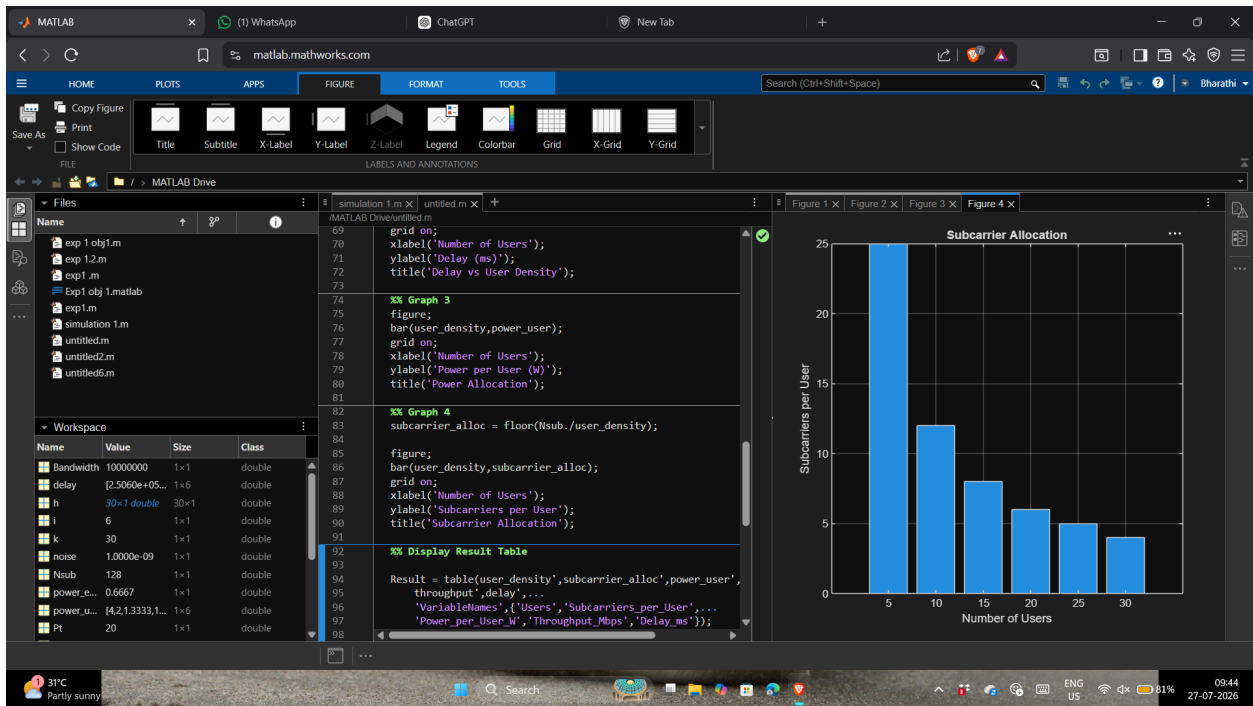
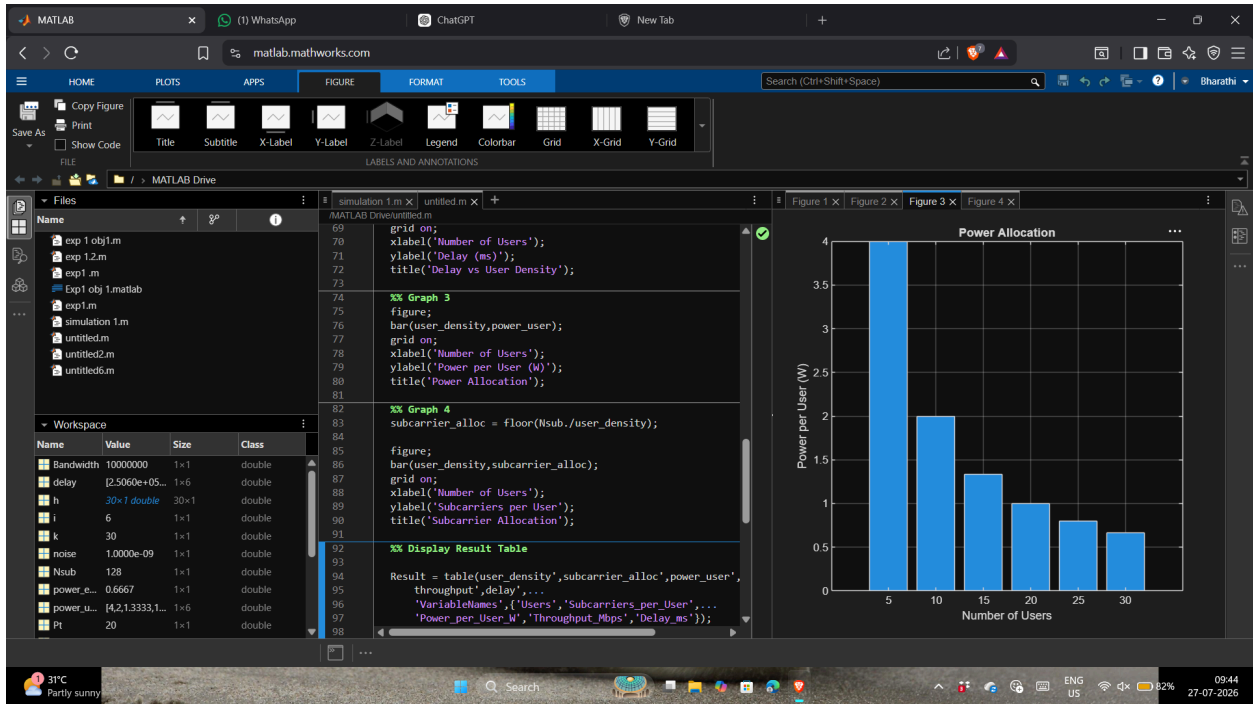
```
clc;
clear;
close all;
%% OFDMA Resource Allocation Simulation
% Parameters
Bandwidth = 10e6;      % 10 MHz
Nsub = 128;            % Number of subcarriers
Pt = 20;               % Total transmit power (Watts)
user_density = [5 10 15 20 25 30];
throughput = zeros(size(user_density));
delay = zeros(size(user_density));
power_user = zeros(size(user_density));
fprintf('\nOFDMA RESOURCE ALLOCATION SIMULATION\n');
fprintf('-----\n');
for i = 1:length(user_density)
    users = user_density(i);
    % Equal Subcarrier Allocation
    sub_per_user = floor(Nsub/users);
    % Equal Power Allocation
    power_each = Pt/users;
    % Random Channel Gain
    h = 0.5 + rand(users,1);
    noise = 1e-9;
    rate = zeros(users,1);
    for k=1:users
        SNR = (power_each*h(k))/noise;
        rate(k) = sub_per_user*log2(1+SNR);
    end
    throughput(i)=sum(rate)/1e6;    % Mbps
    % Delay Model
    delay(i)=1000/throughput(i);    % ms
    power_user(i)=power_each;
    fprintf('Users = %2d | Throughput = %.2f Mbps | Delay = %.2f ms\n',...
        users,throughput(i),delay(i));
end
%% Graph 1
figure;
```

```

plot(user_density,throughput,'-ob','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Throughput (Mbps)');
title('Throughput vs User Density');
%% Graph 2
figure;
plot(user_density,delay,'-sr','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Delay (ms)');
title('Delay vs User Density');
%% Graph 3
figure;
bar(user_density,power_user);
grid on;
xlabel('Number of Users');
ylabel('Power per User (W)');
title('Power Allocation');
%% Graph 4
subcarrier_alloc = floor(Nsub./user_density);
figure;
bar(user_density,subcarrier_alloc);
grid on;
xlabel('Number of Users');
ylabel('Subcarriers per User');
title('Subcarrier Allocation');
%% Display Result Table
Result = table(user_density',subcarrier_alloc',power_user',...
    throughput',delay',...
    'VariableNames',{'Users','Subcarriers_per_User',...
    'Power_per_User_W','Throughput_Mbps','Delay_ms'});
disp(Result);

```





21. Simulation of gNodeB traffic handling under user density, latency, throughput, packet loss, and scheduling

```
clc;
clear;
close all;

%% gNodeB Traffic Handling Simulation

% Parameters
numUsers = [10 20 30 40 50 60]; % User Density
Bandwidth = 100e6; % 100 MHz
TotalCapacity = 1000; % Mbps
Packets = 10000; % Total packets

Throughput = zeros(size(numUsers));
Latency = zeros(size(numUsers));
PacketLoss = zeros(size(numUsers));
ScheduledUsers = zeros(size(numUsers));

fprintf('\n5G gNodeB Traffic Handling Simulation\n');
fprintf('-----\n');

for i = 1:length(numUsers)

    users = numUsers(i);

    %% Round Robin Scheduling
    ScheduledUsers(i) = users;

    %% Capacity Shared Among Users
    userCapacity = TotalCapacity/users;

    %% Random Traffic Demand
    trafficDemand = 8 + rand(users,1)*12; % Mbps

    %% Actual Throughput
    throughputUsers = min(userCapacity,trafficDemand);
```

```

Throughput(i) = sum(throughputUsers);

%% Latency Model
Latency(i) = 2 + (users/10) + rand()*0.5;

%% Packet Loss Model
overload = max(0,sum(trafficDemand)-TotalCapacity);

PacketLoss(i) = (overload/TotalCapacity)*100;

fprintf('Users=%2d Throughput=%.2f Mbps Latency=%.2f ms Packet Loss=%.2f %%\n',...
        users,Throughput(i),Latency(i),PacketLoss(i));

end

%% Graph 1
figure;
plot(numUsers,Throughput,'-ob','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Throughput (Mbps)');
title('Throughput vs User Density');

%% Graph 2
figure;
plot(numUsers,Latency,'-sr','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Latency (ms)');
title('Latency vs User Density');

%% Graph 3
figure;
plot(numUsers,PacketLoss,'-dg','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Packet Loss (%)');
title('Packet Loss vs User Density');

%% Graph 4

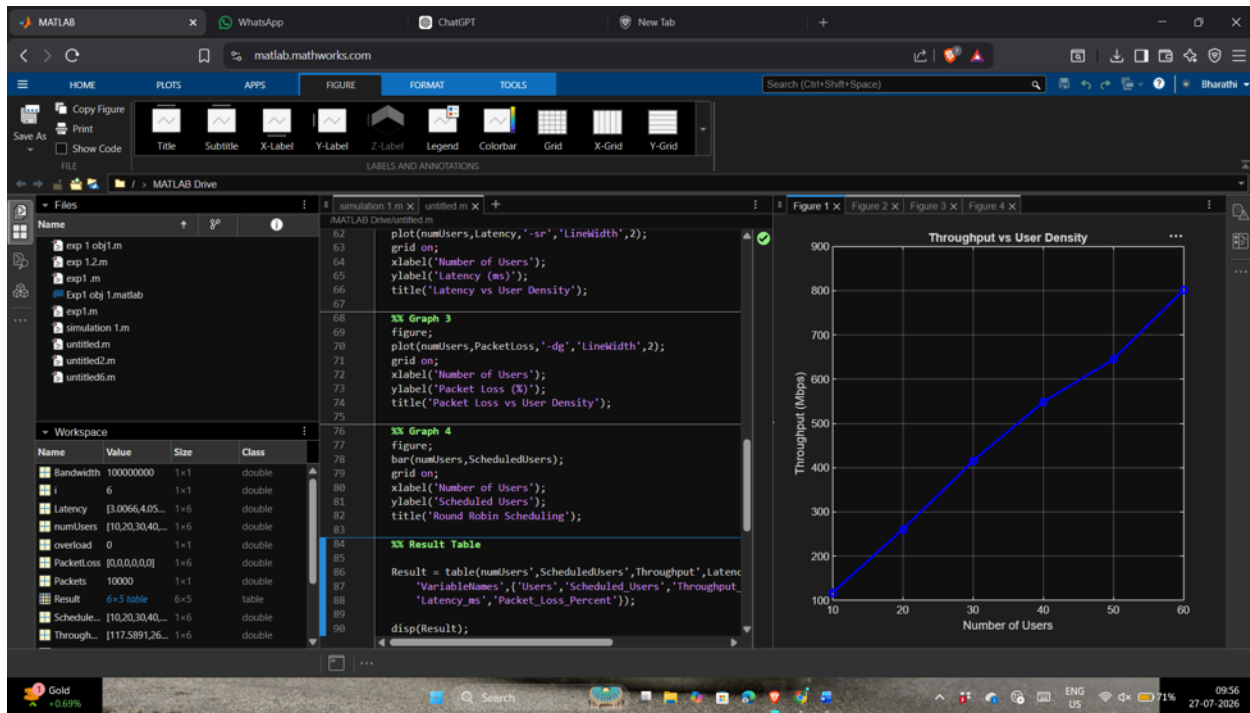
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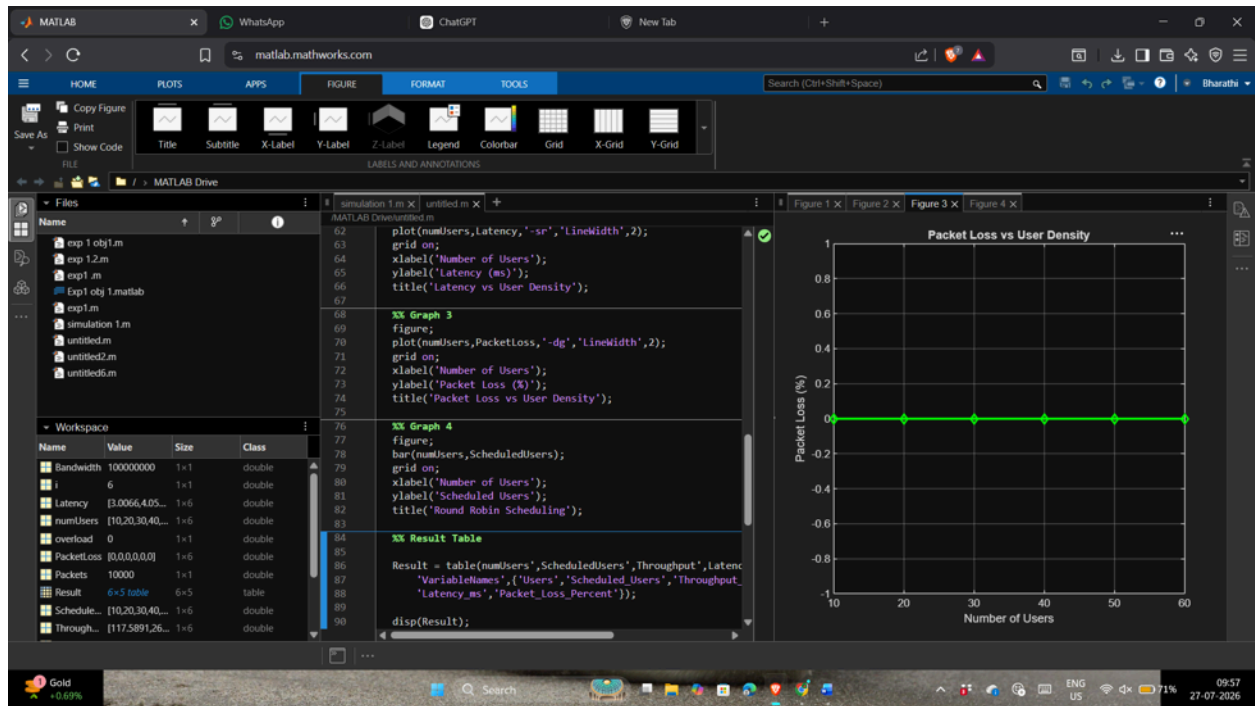
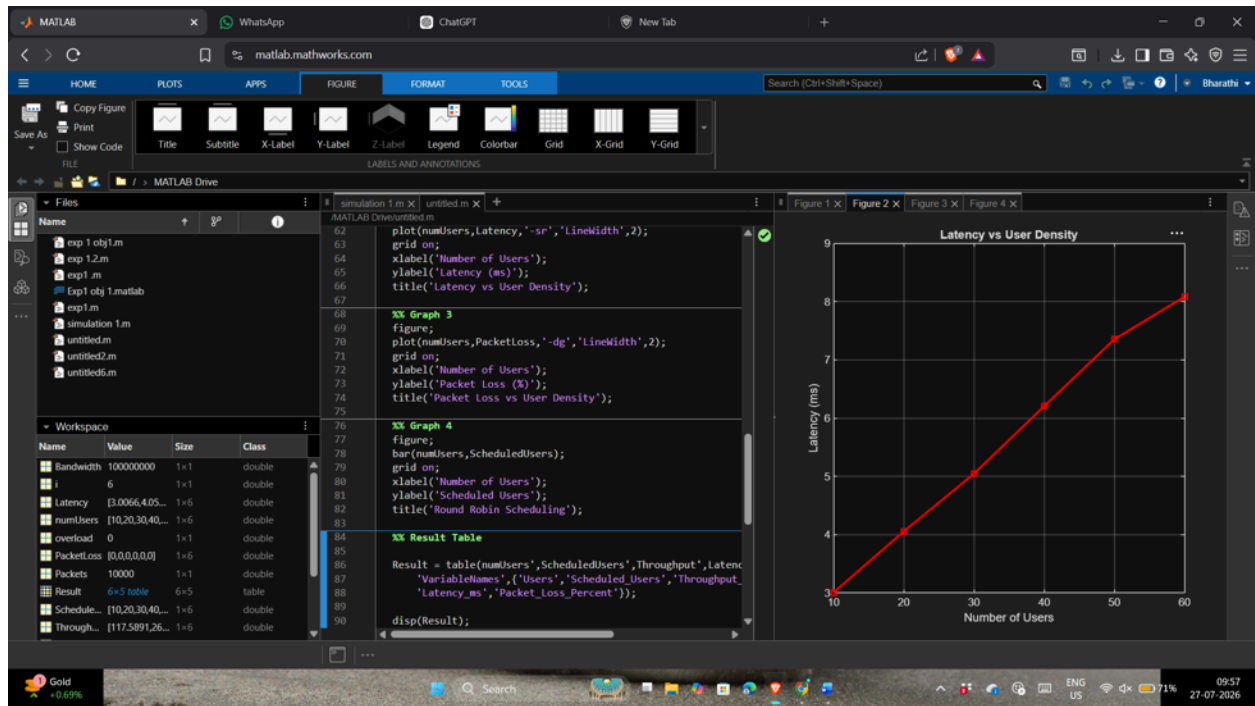
```
figure;
bar(numUsers,ScheduledUsers);
grid on;
xlabel('Number of Users');
ylabel('Scheduled Users');
title('Round Robin Scheduling');
```

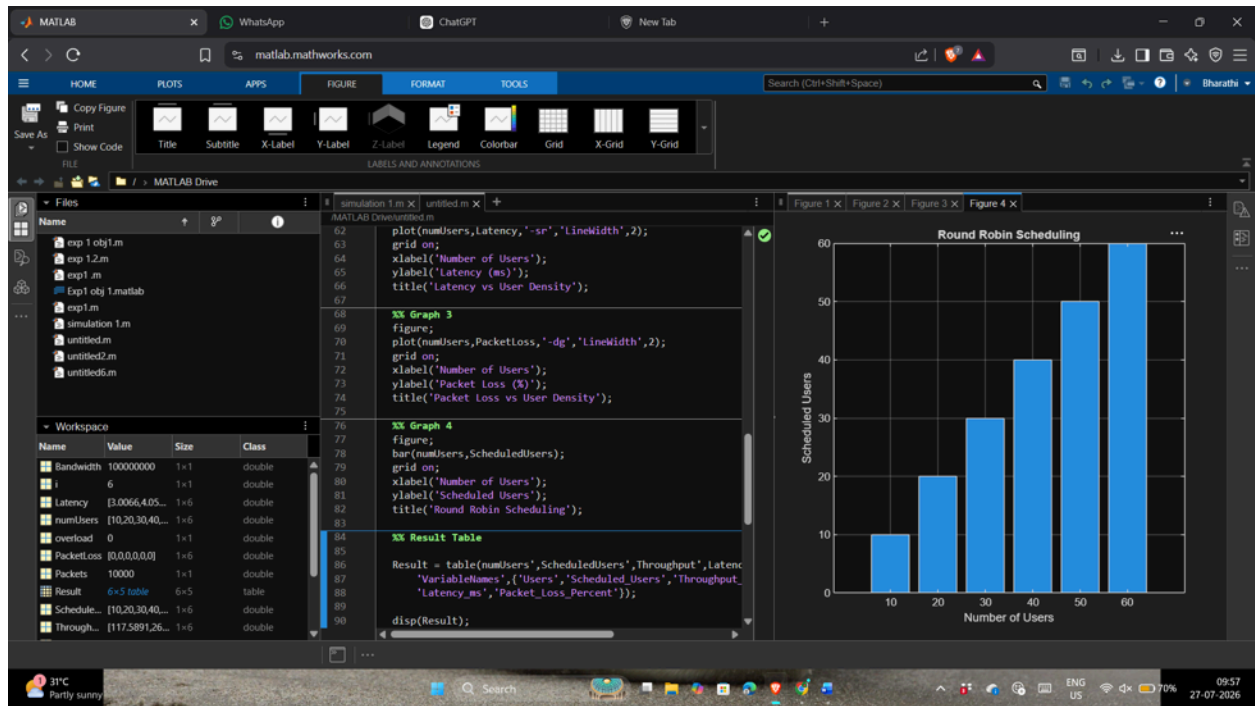
```
%% Result Table
```

```
Result = table(numUsers,'ScheduledUsers','Throughput','Latency','PacketLoss',...
    'VariableNames',{'Users','Scheduled_Users','Throughput_Mbps',...
    'Latency_ms','Packet_Loss_Percent'});
```

```
disp(Result);
```







Code of Conduct:

I bharathi kadiveti(192412132) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bharathi k

Simulation Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation(10)	Shows good understanding with minor gaps(8)	Partial understanding; some misconceptions (6)	Lacks understanding of key concepts(4)
Model Design	25%	Develops accurate, well-structured simulation/model independently (12.5)	Develops correct model with minor errors(10)	Model is incomplete or requires guidance(8)	Unable to develop a proper model(6)
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution(10)	Uses tools with minor errors or guidance(8)	Limited ability to use tools properly(6)	Unable to use simulation tools(4)
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results(10)	Analysis is reasonable with minor gaps(8)	Superficial analysis; limited interpretation(6)	Unable to analyze or interpret results(4)
Documentation	15%	Presents results clearly with proper documentation and structured reporting(7.5)	Documentation is adequate with minor issues(5)	Poorly organized or incomplete documentation (3)	No proper documentation or presentation (1)